



MALAPPURAM SAHODAYA APTITUDE TEST 2024-25 DISTRICT LEVEL

Class: +1 PCM

Date: 04/01/2025

Marks: 100

Name of the Student: :.....

General Instructions:

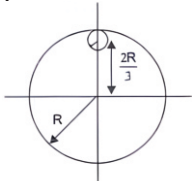
1. Read the questions carefully.
2. Attempt all the questions.
3. No negative marking.
4. Answers to be marked in the OMR sheet. Darken the bubble completely. Partially darkened bubbles will not be considered.
5. Question numbers 1 to 100 carries 1 marks each.

	(Read the passage carefully and answer the questions 1 to 5 choosing suitable options given below:) The old, worn-out piano had been left untouched in the corner of the room for years. Despite its age, the piano still held a certain charm. Its yellowed keys seemed to whisper stories of the past, of laughter and tears, of joy and sorrow. As Emma's fingers danced across the keys, the piano came alive, filling the room with a warm, rich melody that transported her to a different era
1.	Identify the overall tone of the passage A) Melancholic B) Nostalgic C) Critical D) Humorous
2.	Specify the synonym for the word "worn-out" in the passage? A) Old B) Tattered C) Dilapidated D) Vintage
3.	State the effect of Emma's playing on the piano? A) It makes the piano look old. B) It brings back memories of the past. C) It fills the room with a warm melody. D) It makes Emma feel sad.
4.	What can be inferred about Emma's relationship with music? A) She is a professional musician. B) She has a deep emotional connection with music. C) She is learning to play the piano. D) She has no interest in music.
5.	Select a suitable title for the passage from the given options: A) "The Old Piano" B) "A Melody of Memories" C) "Emma's Piano" D) "The Forgotten Instrument"
6.	Fill in the blanks with the correct idiomatic expression: After a week of travelling, Sarah started to _____ and decided to take a day off to rest. A) Burn the midnight oil B) Feel under the weather C) Take a leap of faith D) Beat around the bush
7.	Which of the following is a gerund phrase? A) The dog barks loudly B) She is typing an email C) We went hiking in the mountains D) The baby is sleeping quietly
8.	Choose the correct adjective to complete the sentence: The tourists got lost in the _____ streets of the old city. A) Narrow B) Winding C) Congested D) Deserted
9.	Which of the following is a homophone for "write"? A) right B) rite C) wright D) all of the above

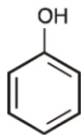
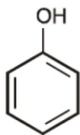
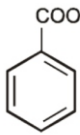
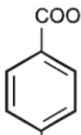
10.	The new policy has made it imperative for employees to attend the training sessions. What does the word "imperative" mean in this sentence? A) Optional B) Desirable C) Urgent D) Mandatory
11.	From the given alternatives, choose the indirect speech of the given sentence, "If you don't keep quiet, I shall shoot you", he said to her in a calm voice. A) He warned her to shoot if she didn't keep quiet calmly. B) He said calmly that I shall shoot you if you don't be quiet. C) He warned her calmly that he would shoot her if she didn't keep quiet. D) Calmly he warned her that be quiet or else he will have to shoot
12.	Identify the literary device in the given sentence and select the option that employs the same device. "The world is a stage." A) "He is a shining light in the darkness." B) "The city streets were a jungle." C) "The sun smiled down on us." D) "The waves crashed against the shore."
13.	Complete the sentence with the correct idiom: "The new policy has been a huge success, and the recent award is just the _____." A) icing on the cake B) tip of the iceberg C) pot of gold D) feather in one's cap
14.	Fill in the blanks using the correct preposition: "The company's decision to downsize was attributed _____ a combination of factors, including economic downturn and increased competition." A) to B) with C) by D) from
15.	Choose the correct form of verb from the given sentence: If she _____ her project on time, she would have received her promotion. A) finished B) finish C) have finished D) had finished
16.	Choose the word that best completes the sentence. The manager's _____ tone during the meeting made everyone feel apprehensive. A) assertive B) sincere C) sarcastic D) polite
17.	Identify the suitable option to form a meaningful sentence The new employee's _____ attitude towards her work was evident in the high quality of her projects, despite the fact that she had only been with the company for a short time. A) apathetic B) androgynous C) enthusiastic D) ambivalent
18.	Fill up using suitable options: The latest policy is _____ the employees' concerns about job security. A) in response to B) in light of C) despite of D) in spite of
19.	Identify the literary device used in the given line, 'The sun came up upon the left, Out of the sea came he' A) Simili B) Metaphor C) Personification D) Hyperbole
20.	Order the following sentences to form a meaningful paragraph P: The owner checked outside and found that there was nothing to worry about. Q: The dog wagged its tail, happy to have done its job of alerting its family to a possible threat. R: The owner scolded the dog, telling him to be quiet. S: The dog barked loudly, waking up everyone in the house A) SRPQ B) PQRS C) PRSQ D) RPSQ

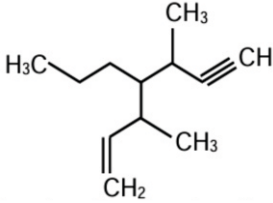
21.	Identify the synonym for the word "APATHY" from the given options: A) Indifference B) Enthusiasm C) Interest D) Passion
22.	The word "OPTIMISM" is best described as having a positive outlook. Which of the following words is the antonym of 'OPTIMISM'? A) Pessimism B) Enthusiasm C) Confidence D) Ambition
23.	Identify the relative clause in the following sentence: "The researcher, who had spent years studying the phenomenon, presented her findings at the conference." A) The researcher presented her findings at the conference B) who had spent years studying the phenomenon C) her findings at the conference D) The researcher had spent years studying the phenomenon
24.	Select the suitable adjective of the following options: The _____ nature of the evidence made it difficult for the detective to crack the case. A) Circumstantial B) Empirical C) Enigmatic D) Perfidious
25.	Select the suitable adverb from the given options: The professor _____ gives extensions on deadlines, so we were surprised when she agreed to our request. A) Rarely B) Occasionally C) Frequently D) Usually
26.	A is the father of C and D is the son of B.E is the brother of A .If C is the sister of D,How is B related to E? A) Daughter B) Brother-in law C) Husband D) Sister-in law
27.	A man walks 5 km toward south and then turns to the right. After walking 3 km he turns to the left and walks 5 km. Now in which direction is he from the starting place? A) North East B) South west C) East D) West
28.	Five people P, Q, R, S and T are sitting in a straight row, facing north. Q sits to the immediate left of S. Only three people sit between R and T. P sits to the immediate left of T. Who sits in the middle of the row? A) P B) Q C) S D) T
29.	A cube painted on all six faces is cut into 64 equal smaller cubes. How many smaller cubes have exactly two faces painted? A) 24 B) 16 C) 12 D) 8
30.	Statements: Some pens are pencils. All pencils are erasers. Conclusions: I Some erasers are pencils. II Some pencils are not pens. Which conclusion follows? A) Only 1 B) Only 2 C) Both 1 and 2 D) Neither 1 nor 2
31.	A and B together can complete a work in 12 days. B and C can complete it in 15 days, and C and A can complete it in 20 days. In how many days will A, B, and C together complete the work? A) 10 B) 12 C) 8 D) 6
32.	A train 200 meters long is running at a speed of 72 km/h. How long will it take to cross a railway signal? A) 10 seconds B) 12.5 seconds C) 15 seconds D) 20 seconds

33.	In a single row, there are some boys and all of them are facing north. Gaurav rank is 29th from the left ends and Tarun rank is 27th from the right end. If 26 students are sitting in between them. Find the number of students in the class. A) 21 B) 9 C) 10 D) 82
34.	What day of the week was 1st January 1900? A) Monday B) Tuesday C) Wednesday D) Sunday
35.	If today is Sunday, what will be the day of the week 500 days from now? A) Friday B) Saturday C) Wednesday D) Monday
36.	What is the angle between the hour and minute hands of a clock at 4:40? A) 100° B) 120° C) 140° D) 160°
37.	At what time between 3 and 4 o'clock will the hands of the clock coincide with each other? A) 166/11 past 3 B) 191/11 past 3 C) 147/11 past 3 D) 180/11 past 3
38.	 A) D B) E C) A D) B
39.	A certain code language, 'PENCIL' is written as 'KLNIPC'. How will 'DOCTOR' be written in that code language? A) CRODOT B) CDOTOR C) CROD TO D) CDOTRO
40.	If a clock shows 2:15, then what time will its image in a plane mirror show? A) 9:45 B) 10:45 C) 11:45 D) 12:45
41.	If 'P' means ' $\div$ ', 'Q' means '+', 'R' means '-', and 'S' means ' $\times$ ', then what is the value of $18 S 12 P 4 Q 5 R 6$? A) 15 B) 19 C) 21 D) 23
42.	Ecology is related to Environment in the same way as Histology is related to A) Fossils B) History C) Tissues D) Hormones
43.	Six members of a family, M, N, O, P, and Q are travelling together. M is the son of N, but N is not the mother of M. L and N are a married couple. P is the brother of N. O is the daughter of L. Q is the brother of M. How many male members are there in the family? A) 4 B) 3 C) 2 D) 1
44.	An old clock is set at 5 am. If the old clock loses 16 minutes in 24 hours, what will be the exact time when the clock indicates 10 pm on the 4th day? A) 10 pm B) 1.15 pm C) 11.30 pm D) 11 pm
45.	By increasing the radius of a sphere by 50 per cent, by how much per cent will the area of its surface increase? A) 100% B) 145% C) 125% D) 150%
46.	40 % is the passing mark. A student gets 250 marks yet fails by 38 marks. What are the maximum marks? A) 800 B) 840 C) 720 D) 750

47.	If the English letters A to Z are written in a reverse order, then what is the fourth letter to the right of 12th letter from the left? A) K B) J C) R D) L
48.	If 40% of a number is equal to two-third of another number, then what is the ratio of the first number to the second number? A) 2:5 B) 2:7 C) 5:3 D) 5:7
49.	If in a certain code BAT=23 and CAT=24, then how will you Code BALL? A) 27 B) 28 C) 32 D) 120
50.	How many triangles are there in the given figure? A) 13 B) 32 C) 21 D) 24
51.	The density of a solid ball is to be determined in an experiment. The diameter of the ball is measured with a screw gauge, whose pitch is 0.5 mm and there are 50 divisions on the circular scale. The reading on the main scale is 2.5 mm and that on the circular scale is 20 divisions. If the measured mass of the ball has a relative error of 2%, the relative percentage error in the density is a) 0.9% b) 2.4% c) 3.1% d) 4.2%
52.	A particle is moving with a uniform speed in a circular orbit of radius R in a central force inversely proportional to the n^{th} power of R . If the period of rotation of the particle is T , then A) $T \propto R^{(n+1)/2+3}$ B) $T \propto R^{3/2}$ C) $T \propto R^{(n/2)+1}$ D) $T \propto R^{n/2}$
53.	A solid sphere of volume V experiences a viscous force F when descending with a speed v in a liquid. If another solid sphere of volume $27V$ descends with the same speed v in the same liquid, it experiences a viscous force A) $12F$ B) $9F$ C) $6F$ D) $3F$
54.	A parachutist after bailing out falls 50 m without friction. When the parachute opens, it decelerates at 2m/s^2 . He reaches the ground with a speed of 3m/s . At what height, did he bailout? A) 293 m B) 111 m C) 182 m D) 91 m
55.	From a uniform circular disc of radius R and mass $9M$, a small disc of radius $\frac{R}{3}$ is removed as shown in the figure. The moment of inertia of the remaining disc about an axis perpendicular to the plane of the disc and passing through centre of disc is :  A) $\frac{37}{9}MR^2$ B) $4MR^2$ C) $\frac{40}{9}MR^2$ D) $10MR^2$
56.	A train is moving along a straight line with a constant acceleration 'a'. A girl standing in the train throws a ball forward with a speed of 10m/s , at an angle of 60° to the horizontal. The girl has to move forward by 1.15 m inside the train to catch the ball back at the initial height. The acceleration of the train, in m/s^2 , is A) 4 m/s^2 B) 8 m/s^2 C) 5 m/s^2 D) 10 m/s^2
57.	A silver atom in a solid oscillates in simple harmonic motion in some direction with a frequency of 10^{12}s^{-1} . What is the force constant of the bonds connecting one atom with the other? (Mole wt. of silver 108 and Avogadro number $6.02 \times 10^{23}\text{ gm mole}^{-1}$) A) 6.4Nm^{-1} B) 7.1Nm^{-1} C) 2.2Nm^{-1} D) 5.5Nm^{-1}

58.	A travelling wave in a medium is given by the equation $y = a \sin(\omega t - kx)$. The maximum acceleration of the particle in the medium is A) $a\omega^2$ B) $a\omega$ C) $k\omega$ D) ω/k
59.	A 20 cm long capillary tube is dipped in water. The water rises up to 8 cm. If the entire arrangement is put in a freely falling elevator the length of the water column in the capillary tube will be A) 20 cm B) 10 cm C) 8 cm D) 4 cm
60.	Three moles of an ideal gas are in a rigid cubical box with sides of length 0.170 m. The ratio of the forces that the gas exerts on each of the six sides of the box when the gas temperature are 27°C and 127°C is A) 6:1 B) 3:1 C) 3:2 D) 3:4
61.	100 g of water is heated from 30°C to 50°C . Ignoring the slight expansion of the water, the change in its internal energy is (specific heat of water is $4184\text{Jkg}^{-1}\text{K}^{-1}$) A) 4.2 kJ B) 84 kJ C) 2.1 kJ D) 8.4 kJ
62.	Three thin wires of equal length are suspended from the top of a roof. The respective ratio of their area of cross section is 1:2:4 and Young's moduli is 4:2:1, then the ratio of their weights to be attached at the other ends to obtain same elongation in them is A) 1:1:1 B) 1:2:4 C) $1:\sqrt{2}:2$ D) $2:\sqrt{2}:1$
63.	Two taps supply water to a container, one at the temperature of 20°C at the rate of 2kg/ minute and another at 80°C at the rate of 1kg/ minute. If the container gets water from the two taps simultaneously for 10 minutes, then the temperature of water in the container is A) 30°C B) 40°C C) 35°C D) 45°C
64.	The potential energy function for the force between two atoms in a diatomic molecule is approximately given by $U(x) = \frac{a}{x^{12}} - \frac{b}{x^6}$ where a and b are constant and x is the distance between the atoms. If the dissociation energy of the molecule is $(U(x = \infty) - U \text{ at equilibrium})$, D is A) $b^2/6a$ B) $b^2/2a$ C) $b^2/4a$ D) $b^2/12a$
65.	Statement 1: A block of mass m starts moving on a rough horizontal surface with a velocity v . It stops due to friction between the block and the surface after moving through a certain distance. The surface is now tilted to an angle of 30° with the horizontal and the same block is made to go up on the surface with the same initial velocity v . The decrease in mechanical energy in the second situation is smaller than that in the first situation. Statement-2: The coefficient of friction between the block and the surface decreases with the increase in the angle of inclination. A) Statement-1 and 2 are true and statement-2 is a correct explanation for statement-1 B) Statement-1 and 2 are true and statement-2 is not a correct explanation for statement-1 C) Statement-1 is true, statement-2 is false D) Statement-1 is false, statement-2 is true.
66.	Volume of 0.6 M NaOH required to neutralize 30cm^3 of 0.4 M HCl is A) 30cm^3 B) 20cm^3 C) 50cm^3 D) 45cm^3
67.	What is the shortest wavelength line in the Paschen series of Li^{2+} ion? A) $\frac{R}{9}$ B) $\frac{9}{R}$ C) $\frac{1}{R}$ D) $\frac{9R}{4}$

68.	B has a smaller first ionization enthalpy than Be. Consider the following statements : (I) It is easier to remove 2 p electron than 2 s electron (II) 2 p electron of B is more shielded from the nucleus by the inner core of electrons than the 2 s electrons of Be (III) 2 s electron has more penetration power than 2 p electron (IV) atomic radius of B is more than Be (atomic number B=5, Be=4) The correct statement are : A) (I), (III) & (IV) B) (II), (III) & (IV) C) (I), (II) & (IV) D) (I), (II) & (III)
69.	The type of hybridisation and number of lone pair(s) of electrons of Xe in XeOF_4, respectively, are A) $\text{sp}^3 \text{d}$ and 2 B) $\text{sp}^3 \text{d}^2$ and 2 C) $\text{sp}^3 \text{d}^2$ and 1 D) $\text{sp}^3 \text{d}$ and 1
70.	Which of the following species is not paramagnetic: A) NO B) CO C) O_2 D) B_2
71.	A system absorbs 600 J of heat and does work equivalent to 300 J on its surroundings. The change in internal energy is A) 300 J B) 400 J C) 500 J D) 600 J
72.	In thermodynamics which one of the following is not an intensive property A) Pressure B) Density C) Volume D) Temperature
73.	For which of the following systems at equilibrium at constant temperature, doubling the volume will cause a shift to the right? A) $\text{H}_2(\text{g}) + \text{Cl}_2(\text{g}) \rightleftharpoons 2\text{HCl}(\text{g})$ B) $2\text{CO}(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{CO}_2(\text{g})$ C) $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$ D) $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3 + \text{Cl}_2(\text{g})$
74.	The solubility of A_2X_3 is $y \text{ mol dm}^{-3}$. Its solubility product is A) $6y^2$ B) $64y^4$ C) $36y^5$ D) $108y^5$
75.	The following equilibrium is established when HClO_4 is dissolved in weak acid HF solvent : $\text{HF} + \text{HClO}_4 \rightleftharpoons \text{ClO}_4^- + \text{H}_2\text{F}^+$ Which of the following is correct set of conjugate acid base pair? A) HF and HClO_4 B) HF and ClO_4^- C) HF and H_2F^+ D) HClO_4 & H_2F^+
76.	Nitric oxide acts as a reducing agent in the reaction A) $4\text{NH}_3 + 5\text{O}_2 \rightarrow 4\text{NO} + 6\text{H}_2\text{O}$ B) $2\text{NO} + 3\text{I}_2 + 4\text{H}_2\text{O} \rightarrow 2\text{NO}_3^- + 6\text{I}^- + 8\text{H}^+$ C) $2\text{NO} + \text{H}_2\text{SO}_3 \rightarrow \text{N}_2\text{O} + \text{H}_2\text{SO}_4$ D) $2\text{NO} + \text{H}_2\text{S} \rightarrow \text{N}_2\text{O} + \text{S} + \text{H}_2\text{O}$
77.	The number of electrons lost or gained during the change $\text{Fe} + \text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + \text{H}_2$ A) 2 B) 4 C) 6 D) 8
78.	The correct acidity order of the following is: A) (III) > (IV) > (II) > (I) B) (IV) > (III) > (I) > (II) C) (III) > (II) > (I) > (IV) D) (III) > (IV) > (II) > (I)  (I)  (II)  (III)  (IV)

79.	Racemic mixture is formed by mixing two: A) Isomeric compounds B) Chiral compounds C) Meso compounds D) Optical isomers
80.	The IUPAC name of the following compound is: A) 3-methyl-4-(3-methylprop-1-enyl)-1-heptyne B) 3,5-dimethyl-4-propylhept-1-en-6-yne C) 3,5-dimethyl-4-propylhept-6-en-1-yne D) 3-methyl-4-(1-methylprop-2-ynyl)-1-heptene 
81.	In a certain town, 25% families own a cell phone, 15% families own a scooter and 65% families own neither a cell phone nor a scooter. If 1500 families own both a cell phone and a scooter, then the total number of families in the town is A) 10000 B) 20000 C) 30000 D) 40000
82.	The domain of definition of the function $f(x) = \sqrt{1 + \log_e(1 - x)}$ is A) $-\infty < x \leq 0$ B) $-\infty < x \leq \frac{e-1}{e}$ C) $-\infty < x \leq 1$ D) $x \geq 1 - e$
83.	If $5 \cos 2\theta + 2 \cos^2 \frac{\theta}{2} + 1 = 0$, where $0 < \theta < \pi$, then the values of θ are A) $\frac{\pi}{3} \pm \pi$ B) $\frac{\pi}{3}, \cos^{-1}\left(\frac{3}{5}\right)$ C) $\cos^{-1}\left(\frac{3}{5}\right) \pm \pi$ D) $\frac{\pi}{3}, \pi - \cos^{-1}\left(\frac{3}{5}\right)$
84.	If $\frac{\sin(x+y)}{\sin(x-y)} = \frac{a+b}{a-b}$, then $\frac{\tan x}{\tan y}$ is equal to A) $\frac{a^2}{b^2}$ B) $\frac{a}{b}$ C) $\frac{b}{a}$ D) $\frac{a^2+b^2}{a^2-b^2}$
85.	Using mathematical induction, the numbers a_n's are defined by $a_0 = 1, a_{n+1} = 3n^2 + n + a_n, (n \geq 0)$. Then, an is equal to A) $n^3 + n^2 + 1$ B) $n^3 - n^2 + 1$ C) $n^3 - n^2$ D) $n^3 + n^2$
86.	If $a = e^{i\frac{2\pi}{3}}$, then the equation whose roots are $a+a-2$ and a^2+a-4 is A) $x^2 - 2x + 4 = 0$ B) $x^2 + x + 4 = 0$ C) $x^2 + 2x - 4 = 0$ D) $x^2 + 2x + 4 = 0$
87.	The value of x such that $3^{2x} - 2(3^{x+2}) + 81 = 0$, is A) 1 B) 2 C) 3 D) 4
88.	If $x^2 + 4ax + 2 > 0$ for all values of x, then a lies in the interval A) $(-2, 4)$ B) $(1, 2)$ C) $(-\sqrt{2}, \sqrt{2})$ D) $\left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$
89.	Let A and B be two sets containing 2 elements and 4 elements, respectively. The number of subsets of $A \times B$ having 3 or more elements, is A) 256 B) 220 C) 219 D) 211
90.	If the coefficients of x^2 and x^3 in the expansion of $(3+ax)^9$ is same, then the value of a is A) $\frac{3}{7}$ B) $\frac{7}{3}$ C) $\frac{7}{9}$ D) $\frac{9}{7}$

91.	A student read common difference of an AP as -3 instead of 3 and obtained the sum of first 10 terms as -30. Then, the actual sum of first 10 terms is equal to A) 240 B) 120 C) 300 D) 180
92.	The straight lines $x+y=0$, $5x+y=4$ and $x+5y=4$ form A) An isosceles triangle B) An equilateral triangle C) A scalene triangle D) A right angled triangle
93.	A line passes through the point of intersection of the lines $100x+50y-1=0$ and $75x+25y+3=0$ and makes equal intercept on the axes. Its equation is A) $25x+25y-1=0$ B) $5x-5y+3=0$ C) $25x+25y-4=0$ D) $5x-5y+7=0$
94.	Let the equation of an ellipse be $\frac{x^2}{144} + \frac{y^2}{25} = 1$. Then, the radius of the circle with centre (0, $\sqrt{2}$) and passing through the foci of the ellipse is A). 9 B) 7 C) 11 D) 5
95.	The angle between the lines whose direction cosines satisfy equations $l+m+n=0$ and $l^2=m^2+n^2$, is A) $\frac{\pi}{3}$ B) $\frac{\pi}{4}$ C) $\frac{\pi}{6}$ D) $\frac{\pi}{2}$
96.	If $f(x)=(x^5-1)(x^3+1)$, $g(x)=(x^2-1)(x^2-x+1)$ and $h(x)$ be such that $f(x)=g(x) h(x)$, then $\lim_{x \rightarrow 1} \frac{f(x)}{h(x)}$ is equal to A) 0 B) 5 C) 4 D) 1
97.	If $f(x)=x^2+bx+7$. If $f'(5)=2f'(7/2)$, then the value of b is A) 4 B) 3 C) -4 D) -3
98.	The statement $\sim(p \rightarrow q)$ is equivalent to A) $p(\sim q)$ B) $\sim pq$ C) pq D) $\sim p \sim q$
99.	The AM of 9 terms is 15. If one more term is added to this series, then the AM becomes 16. The value of the added term is A) 30 B) 27 C) 25 D) 20
100.	A poker hand consists of 5 cards drawn at random from a well-shuffled pack of 52 cards. Then, the probability that a poker hand consists of a pair and a triple of equal face values (e.g. 2 sevens and 3 kings or aces and 3 queens, etc.) is A) $\frac{6}{4165}$ B) $\frac{23}{4165}$ C) $\frac{1797}{4165}$ D) $\frac{1}{4165}$
